Biosense Webster
PART OF THE Johnson & Johnson FAMILY OF COMPANIES

VARIPULSE™ Bi-Directional Catheter

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Symbol Definitions

MD
Medical Device

STERILE EO
Sterilized using ethylene oxide

REF
Catalog Number

LOT
Batch Code

UDI
Unique Device Identifier


Manufacturer


Date of manufacture


Do not re-use


Packaging unit


Single sterile barrier system with protective packaging inside


Use-by date


Caution


Consult instructions for use or consult electronic instructions for use


Do not use if package is damaged


Do not use if package is opened


Keep away from sunlight


Keep dry


Temperature limit



MR Unsafe



Caution: Federal (USA) law restricts this device to sale by or on the order of a licensed healthcare practitioner.



The irrigation tubing contains substance N-methyl-2-pyrrolidone, CAS# 872-50-4, in concentration above 0.1% weight by weight. Refer to the ECHA website for more information: https://echa.europa.eu/home.



Pin connector



Electrodes



Spacing



Navigation catheter



Compatible with CARTO™ 3 EP Navigation System



Assembled in Mexico

VARIPULSE™ Bi-Directional Catheter

Caution: Federal (USA) law restricts this device to sale by or on the order of a licensed healthcare practitioner.

- STERILE. Sterilized using ethylene oxide.
- For single use only. Do not resterilize.
- Do not use if the package is opened or damaged.

DEVICE DESCRIPTION

The VARIPULSE™ Bi-Directional Catheter is a steerable, multi-electrode, irrigated, catheter. Attached to the distal end of the shaft is a bi-directional deflectable segment. Attached to the segment is a circular tip containing 10 ring electrodes. The usable length of the catheter is 115 cm.

The catheter was designed to facilitate cardiac electrophysiological mapping (stimulating and recording) and ablation. The catheter was also designed to be used with a Biosense Webster TRUPULSE™ Generator to transmit biphasic high voltage pulses to the electrodes on the catheter to perform ablation.

The circular tip of the catheter is a loop that can be expanded and contracted to accommodate the anatomy of the left atrium. The uncontracted loop diameter is 34 ± 1 mm and the contracted loop diameter is ≤ 25 mm at room temperature with further contraction expected during clinical use due to body temperature. The 10 electrodes on the loop are perforated with a series of holes to facilitate irrigation of the ablation area. The electrodes are made from noble metals and may be used for visualization, stimulation, recording, and ablation.

A standard Luer fitting is attached to the lumen that extends from the handle of the catheter. Heparinized normal saline is injected through the port in the Luer fitting. During mapping and ablation, the saline passes through the internal lumen of the catheter and through the holes of the electrodes. An irrigation pump controls the irrigation flow.

In the loop of the catheter are 3 single axis sensors (SAS) which transmit location information to the CARTO™ 3 System for catheter visualization. Refer to the instructions for use for the CARTO™ 3 System for information on using the catheter for mapping procedures.

INDICATIONS FOR USE

The VARIPULSE™ Catheter is indicated for use in catheter based cardiac electrophysiological mapping (stimulating and recording) and, when used with a TRUPULSE™ Generator, for the treatment of drug refractory recurrent symptomatic paroxysmal atrial fibrillation. The catheter provides location information when used with the CARTO™ 3 System. The device is intended for adults 22 years old and above.

CONTRAINDICATIONS

1. Do not use the catheter if the patient has had an atriotomy within the preceding eight weeks. Recent surgery may increase the risk of perforation.
2. Do not use the catheter in a patient with a myxoma or an intracardiac thrombus. Doing so may precipitate emboli.
3. Do not use the catheter in a patient with a prosthetic valve. Doing so may damage the prosthetic valve, result in device entrapment or embolization of device components.
4. Do not use the catheter in the coronary arterial vasculature. Doing so may damage the coronary arterial vasculature or cause an infarction.
5. Do not use the catheter in a patient with an active systemic infection. Doing so may increase the risk of cardiac infection.
6. Do not use the catheter via the transseptal approach in a patient with an interatrial baffle or patch. The opening may persist and produce an iatrogenic atrial shunt.
7. Do not use the catheter via the retrograde trans-aortic approach. Doing so may increase the risk of perforation.
8. Do not use the catheter in the ventricles. The catheter may become tangled with the valves.

9. Do not use the catheter in a patient with a recent history of myocardial infarction. The patient may be susceptible to ventricular tachycardia.
10. Do not use the catheter in a pregnant woman. The effects of pulse field energy on a fetus are unknown.
11. Do not use the catheter in arterial or jugular access sites. Doing so may cause injury to the vasculature and may increase incidents of hematoma.

PATIENT TARGET GROUP

This device is targeted at patients who have been diagnosed with cardiac arrythmias and are undergoing an electrophysiology procedure.

WARNINGS AND PRECAUTIONS

1. The safety and efficacy of the VARIPULSE™ Catheter when used with a TRUPULSE™ Generator, has been determined in the admIRE trial in the treatment of subjects with symptomatic paroxysmal atrial fibrillation (PAF). The safety and effectiveness of the device use in the treatment of other arrhythmias including persistent atrial fibrillation have not been established.
2. The safety and effective use of this device for ablations beyond pulmonary vein isolation for the treatment of atrial fibrillation has not been established and may increase the risk of patient injury such as peri-procedural stroke and transient ischemic attack.
3. The safety and effectiveness of catheter ablation for the treatment of atrial fibrillation in patients with significant left ventricular dysfunction, advanced heart failure, substantial left atrial enlargement, and structural heart disease have not been established.
4. The long-term risks of the application of PF ablation in the heart have not been established.
5. Because of the potential effects of PF energy on the fetus are unknown, use of the catheter in pregnant women should be avoided.
6. Some patients may have impaired ability to handle the fluid volume delivered during irrigation. Fluid overload may lead to pulmonary edema or heart failure during or after the procedure. Elderly patients and patients with congestive heart failure or renal insufficiency are particularly susceptible. Before the procedure, identify the patient's risk of fluid overload. Monitor the patient's fluid volume throughout the procedure to avoid fluid overload.
7. A patient who has had a prior ablation procedure may be at greater risk for perforation and/or pericardial effusion with the use of the catheter.
8. Cardiac ablation procedures must be performed by appropriately trained personnel in a fully equipped electrophysiology laboratory that meets the requirements of a medical room, as specified in the IEC 60601-1 standard (Medical electrical equipment – Part 1: General requirements for basic safety and essential performance).
9. Only use the catheter in a fully equipped catheter or electrophysiological laboratory. It is recommended that the laboratory should have its corresponding equipment which may include but not limited to fluoroscopy, pacing, recording, defibrillator or intracardiac echocardiography units.
10. The catheter must be used only with a Biosense Webster TRUPULSE™ Generator and the Biosense Webster interface cable for multi-electrode ablation catheters. Before using these devices and compatible accessories, completely read and understand the applicable instructions for use.
11. Do not attempt to operate the catheter prior to completely reading and understanding the applicable instructions for use.
12. Ensure that the catheter is the only metal in the left atria during energy application. Metal from other electrophysiology (EP) devices in left atrium can disrupt the electric field and potentially lead to injury.
13. Throughout the procedure, do not allow the patient to come in contact with metal surfaces such as the operating table supports. Doing so may cause injury or death of the patient by electrocution.
14. Metals from adjacent structure during energy application such as other electrophysiology (EP) devices, probes (e.g., non-insulated esophageal temperature probe), or implants can disrupt the electric field and potentially lead to injury.
15. When using the catheter with other devices, maintain proper connections with associated interface cables. Improper connections may lead to patient harm.

16. Do not allow the connectors on the catheter and on the interface cables to contact parts that are conductive (such as mains outlets or metallic objects). Doing so may cause injury or death of the patient by electrocution.
17. Improper handling, connection, or use of the catheter may result in injury to the patient or operator.
18. Electrodes and probes used for monitoring and stimulating the heart can provide paths for high frequency current. To reduce the risk of burns, place these electrodes and probes as far as possible from the ablation site. Protective impedances may also reduce the risk of burns and permit continuous monitoring of the electrocardiogram during the application of PF ablation.
19. Implantable pacemakers, implantable cardioverter/defibrillators (ICDs), and metal stents can disrupt the electric field produced by PF ablation.
20. Pay close attention when applying Pulse Field (PF) ablation in patients with implantable pacemakers and implantable cardioverter/defibrillator (ICDs). Implantable pacemakers and ICDs may be adversely affected by PF energy. Safety and Effectiveness have not been evaluated for patients with pacemakers and ICDs. It is important to have temporary external sources of pacing and defibrillation available during ablation and to consider temporarily reprograming the pacing system to minimum output or OFF mode to minimize the risk of inappropriate pacing. Exercise extreme caution during ablation when in close proximity to atrial or ventricular permanent leads; program the ICD to the OFF mode during the ablation procedure; perform complete analysis after ablation.
21. Throughout the procedure, carefully monitor the patient for signs of arrythmia and vagal response. Profound vagal responses have been reported immediately after ablation with irreversible electroporation. Follow standard of care management if the patient experiences a vagal response.
22. Pulsed field ablation may result in an extended pause in cardiac rhythm. Prior to delivering pulsed field (PF) energy, ensure adequate preparations are in place to commence emergency ventricular pacing, administer cardiopulmonary resuscitation, or apply other measures consistent with standard of care should the need arise.
23. Carefully follow the catheter directions for use. In clinical studies with VARIPULSE™ (admIRE[1] and inspIRE[2]) the median number of ablations delivered was 23 (69 applications) and 16 (49 applications) respectively. Device safety and effectiveness were evaluated in all treated subjects, 75% of whom had fewer than 28 ablations (84 applications) delivered in the admIRE clinical study and 17 ablations (52 applications) in the inspIRE clinical study.
24. The risk of patient injury including peri-procedural stroke and transient ischemic attack and product malfunction may increase with an increased number of ablations (energy applications) delivered during the procedure.
25. The safety of discontinuing anticoagulation therapy following catheter ablation of atrial fibrillation has not been established. Anticoagulation therapy in such patients should be administered in accordance with the 2023 ACC/AHA/ACCP/HRS Guideline for the Diagnosis and Management of Atrial Fibrillation.
26. Maintain appropriate levels of recommended anticoagulation therapy before and after the procedure. The safety of discontinuing anticoagulation therapy following catheter ablation of atrial fibrillation has not been established. Anticoagulation therapy in such patients should be administered in accordance with the 2023 ACC/AHA/ACCP/HRS Guideline for the Diagnosis and Management of Atrial Fibrillation.
27. If there is uncertainty regarding the patient's anticoagulation status or rhythm prior to the procedure, evaluate the need to perform a Transesophageal Echocardiogram (TEE) prior to the procedure to confirm absence of mural thrombus and/or thrombus in the left atrial appendage.

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[1] Reddy V, Calkins H, Mansour M, et al. Pulsed field ablation to treat paroxysmal atrial fibrillation: safety and effectiveness in the admIRE pivotal trial. Circulation. Published online September 11, 2024. doi: 10.1161/CIRCULATIONAHA.124.070333.

[2] Duytschaever M, De Potter T, Grimaldi M, et al. Paroxysmal AF ablation using a novel variable-loop biphasic pulsed field ablation catheter integrated with a 3D mapping system: 1-year outcomes of the multicenter inspIRE study. Circ Arrhythm Electrophysiol. 2023;16(3):e01178.

28. Confirm ACT ≥ 350 seconds prior to ablation with the catheter and check every 15-30 minutes while the catheter is in the left atrium. Failure to maintain appropriate anticoagulation levels may increase the risk for thromboembolic events.
29. Follow clinical guidelines and common practice to minimize the risk of thromboembolism. Maintain a constant flow of heparinized normal saline throughout the procedure.
30. Use only the irrigation tubing set recommended for use with the irrigation pump. Using an incompatible irrigation tubing set may compromise the irrigation flow. Do not reuse the irrigation tubing set or use extension tubing. Doing so may compromise the irrigation flow and structural integrity of the tubing and/or may lead to patient harm.
31. Before inserting the catheter in the patient, flush the catheter and irrigation tubing set with heparinized normal saline to remove air bubbles. Air bubbles may cause air embolism and resulting clinical sequelae.
32. Prior to each application of PF ablation, verify continuous irrigation flow by observing the dripping in the tubing drip chamber to confirm saline flow to the ablation electrodes. If irrigation blockage is suspected, remove and inspect the catheter following the instructions in the “Directions for Use” section of this document.
33. Only use compatible deflectable guiding sheath (minimum, 8.5 F inner lumen). A compatible fixed guiding sheath may also be used. Using incompatible guiding sheath may compromise the structural integrity of the catheter, which may lead to patient harm or catheter malfunction.
34. Do not use a short sheath; use of a short sheath may lead to device entrapment, or cardiac perforation.
35. To avoid damaging the loop or the electrodes, do not twist or contract the loop when inserting the catheter into the guiding sheath or removing the catheter from the guiding sheath. If resistance is encountered, do not use excessive force to advance or withdraw the catheter through the guiding sheath.
36. To prevent patient movement during pulsed field (PF) ablation, patient comfort should be maintained through local standards of care for pain management.
37. Catheter advancement should be done under direct imaging guidance (such as fluoroscopy, or intracardiac echocardiography, or with the CARTO™ 3 System). Careful catheter manipulation must be performed in order to avoid cardiac damage, perforation, or tamponade, or entanglement which may lead to valvular damage.
38. Use both direct imaging guidance (such as fluoroscopy and/or intracardiac echocardiography) and electrogram data to monitor catheter advancement and reduce risk of tissue injury.
39. When using the catheter with conventional systems (such as fluoroscopy or intracardiac echocardiography), with the CARTO™ 3 System, careful catheter manipulation must be performed in order to avoid cardiac damage, perforation, or tamponade. Catheter advancement should be done under direct imaging guidance.
40. Do not use excessive force to advance or withdraw the catheter when resistance is encountered. The firmness of the braided tip dictates that care must be taken to prevent perforation of the heart.
41. Minimize or eliminate X-ray exposure during the procedure. Catheter ablation procedures using fluoroscopic imaging may present the potential for significant X-ray exposure, which can result in acute radiation injury as well as increased risk for somatic and genetic effects to both patients and laboratory staff due to the X-ray beam intensity and duration of fluoroscopic imaging. Catheter ablation, when using fluoroscopic imaging, should only be performed after adequate attention has been given to the potential radiation exposure associated with the procedure, and steps have been taken to minimize this exposure. Careful consideration must therefore be given for the use of the catheter in prepubescent children.
42. Unsheath the catheter from the sheath in the middle of the chamber to prevent entrapment or perforation.
43. When maneuvering the catheter in the patient, rotate the catheter clockwise to reduce the risk of entrapping the loop of the catheter in the heart or perforating tissue. To avoid potential damage to anatomical structures, do not attempt to pull the catheter or withdraw it into the guiding sheath when the loop is contracted, or the tip is deflected.

44. During mapping procedures, ensure that the loop of the catheter is fully contracted to reduce the risk of catheter loop entrapment in the heart.
45. When using the catheter in or around the atrioventricular valve, take precautions to prevent the catheter from slipping into the ventricles and becoming entangled with the valves. Entanglement may lead to damage to the valve, entrapment of the device, and/or need for surgical intervention.
46. When ablating near adjacent electrophysiologic structures (sinoatrial node and atrioventricular node) and anatomical structures, take precautions according to current standard of care to minimize collateral damage to the adjacent structures.
47. Do not use pulse field ablation in close proximity to the atrioventricular node. Use caution when delivering PF energy in close proximity to normal conduction system.
48. Failure of the equipment to function correctly at normal settings may indicate failure of an electrical lead. Check the catheter for visible defects before applying PF ablation.
49. The sterile packaging and catheter should be inspected prior to use. Do not use if the packaging or catheter appears damaged.
50. Do not expose the catheter to volatile organic solvents such as alcohol. Doing so may compromise the structural integrity of the device and/or lead to device failure.
51. Do not immerse the catheter handle or the cable connector in fluids; electrical performance may be affected.
52. Do not store or use flammable materials in the surgical area.
53. Do not use the catheter near MRI equipment or magnetic navigation systems because the equipment may induce movement of the catheter, which may cause injury or perforation.
54. The irrigation tubing in the catheter contains N-methyl-2-pyrrolidone (NMP), CAS# 872-50-4, at a level that is above the reporting threshold of 0.1% weight of NMP by weight of irrigation tubing. A toxicology risk assessment concluded that the level of NMP released during catheter ablation procedures is insignificant for adults, pregnant or breastfeeding women, and children.

MRI SAFETY INFORMATION

The catheter is MR Unsafe. Keep it outside the MRI scanner room. The MRI safety of the catheter has not been assessed. Do not use the catheter near MRI equipment because movement or heating of the catheter may occur and the image on the display may become distorted.

STORAGE

Store in the original carton box. Refer to the product label for recommended storage conditions.

STERILIZATION AND USE-BY DATE

The catheter was sterilized using ethylene oxide. Do not use the catheter if the package is opened or damaged. Product and package testing was conducted to support the use-by date printed on the product label. Do not use the catheter after the date on the product label.

The catheter was packaged and sterilized for single use only. Do not reuse, reprocess, or resterilize. Reuse, reprocessing, or resterilization may compromise the structural integrity of the device and/or lead to device failure that in turn may result in patient injury, illness, or death. Also, reprocessing or resterilization of single use devices may create a risk of contamination and/or cause patient infection or cross-infection, including, but not limited to, the transmission of infectious disease(s) from one patient to another, sepsis, and endocarditis. Contamination of the device may lead to injury, illness, or death of the patient.

DISPOSAL

Recycle components or dispose of the product and residual elements or waste items, in accordance with local laws and regulations. Dispose of contaminated product in accordance with standard hospital procedures.

DIRECTIONS FOR USE

To perform ablation, the catheter must be connected to a TRUPULSE™ Generator and to a compatible irrigation pump. For directions on connecting and operating the catheter with these devices, refer to the applicable instructions for use.

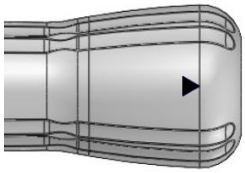
1. Use only a compatible deflectable guiding sheath (minimum 8.5 F inner lumen) or compatible fixed sheath. Using a deflectable sheath is recommended. For compatible guiding sheaths, contact Customer Support or your Biosense Webster representative.
2. Using standard aseptic technique, remove the catheter from the package. Visually verify that the catheter is not damaged. Place the catheter in a sterile work area.
3. Use the Rocker Lever to deflect the catheter tip. When the lever is turned out of the neutral position, the tip deflects in the same direction as the lever. The amount of deflection is relative to the amount of lever rotation. To straighten the tip, return the Rocker Lever to neutral position.
4. Verify that the loop on the catheter tip contracts and expands properly. To contract the loop, point the catheter away from the user and, with the Rotating Contraction Knob on the handle facing the user, turn the Rotating Contraction Knob clockwise until the minimum diameter of the loop is reached. To expand the loop, turn the Rotating Contraction Knob counterclockwise until the maximum diameter of the loop is reached.

Rocker Lever (for Deflection) **Rotating Contraction Knob (for Loop Diameter)**





5. Connect the irrigation tubing set to the Luer fitting on the catheter. The stopcock that is provided with the tubing may be used. Using standard aseptic technique, connect the irrigation tubing set to a saline bag that contains heparinized normal saline (1 IU/ml minimum).
6. Ensure the pump is set to deliver a flow rate of 4 ml/min at idle.
7. Using standard aseptic technique, flush the catheter and irrigation tubing set to remove air bubbles. To verify patency throughout the catheter, hold the catheter with the loop facing downward and parallel to the floor, flush the catheter, and verify that saline flows from each electrode. Hold the catheter with the loop facing upward and use standard aseptic technique to flush air bubbles from the catheter and the tubing.
8. Connect the catheter to the TRUPULSE™ Generator using the Biosense Webster interface cable for multi-electrode ablation catheters as indicated by the matching markings on the ends of the cable and catheter handle.



9. Using standard aseptic technique, flush the guiding sheath to remove air bubbles. Then establish vascular access in a large central vein and insert the guiding sheath. Refer to the instructions for use for the guiding sheath.
10. Before inserting the catheter in the guiding sheath, ensure that the tip section is not deflected by turning the Rocker Lever to the neutral position. To ensure minimal tension on the loop, turn the Rotating Contraction Knob counterclockwise until the maximum diameter of the loop is reached. Start catheter irrigation.
11. Applying slow and gentle force, introduce the catheter through the hemostatic valve on the guiding sheath. Continually irrigate the guiding sheath to ensure that no air is introduced. Carefully advance the catheter into the body.

12. Under direct imaging guidance, carefully advance the catheter into the left atrium.
13. Catheter rotation should always be in the clockwise direction.
14. If the overlap of ablation electrodes cannot be avoided turn off the overlapped electrodes.
15. While performing an ablation do not move the catheter until after 1 complete ablation have been delivered.
16. Before retracting the catheter into the guiding sheath, turn the Rocker Lever to the neutral position to ensure that the tip section is not deflected and turn the Rotating Contraction Knob counterclockwise to ensure that the loop is fully expanded.

Performing PF Ablation with the Catheter

- Confirm ACT ≥ 350 seconds prior to ablation with the catheter, and check every 15-30 minutes while the catheter is in the left atrium.
- Select a transeptal location that will provide reach to all four veins, such as anterior, inferior positions.
- It is recommended to only perform ablation when contact has been confirmed by surrogate markers via ICE, fluoroscopy, electrogram signals, and Tissue Proximity Indicator (TPI) on the CARTO™ map.
- For each ablation, use all 10 electrodes unless there is overlap.
- Do not allow ablation electrodes to overlap during the application of PF energy. Overlapping electrodes will be deselected by the generator. Consider adjusting the catheter position to ensure that 10 electrodes are selected whenever possible.
- It is recommended to ensure diaphragmatic capture before and after ablation while ablating in the region of the right-sided PVs in proximity to the phrenic nerve.

Workflow for Pulmonary Vein Isolation

Table 1 – VARIPULSE™ Parameters

Parameter	VARIPULSE™	VARIPULSE™ Plus	VARIPULSE™ Pro
TRUPULSE™ Generator Software Version	2.2.2 and below	2.2.3	2.2.4 and above
Idle Irrigation Flow rate (ml/min)	4	4	4
Ablation Irrigation Flow rate (ml/min)	4 or 30*[3]	4 or 30[3]	30[3]
Application Delivered during 1 ablation	3	3	4
Inter Application Delay (sec)	10	10	1

*Adjusted manually

- For each pulmonary vein, create a concentric ring of lesion using 4 ablations (2 ostial with the contracted loop and 2 antral with an open loop):
    - The catheter position should be co-axial to the vein.
    - Place an electrode catheter in the right ventricle or a ventricular branch of the coronary sinus to be ready for ventricular pacing in the event of vagal response.
- Perform 1 ostial ablation with the contracted loop without moving the catheter.
- Within the ostial area (with the contracted loop), reposition the catheter and perform an additional ablation. The new catheter position should cover any potential gaps from the first ostial catheter position.

[3] Based on preclinical data, 30 ml/min flow rate reduces electrode surface temperature and may reduce char and thrombus formation. However, there is no clinical evidence that 30 ml/min flow rate reduces risk of stroke/TIA.

- Ensure that all applications are complete (1 complete ablation) (generator screen reads 100%).
    - If any ablation is not complete (less than 100%), then repeat the incomplete ablation in the same location.
- After completing the 2 ostial ablations (with the contracted loop), move the catheter into an antral side (with an open loop) and follow a similar workflow completing 1 ablation at the same catheter position and repositioning to complete the final antral ablation.
- Deliver 4 ablations per vein.
- In anatomy with a common PV, deliver 8 ablations per common PV.
- If isolation is not obtained, apply additional applications of PF ablation where needed.
    - Reposition the catheter between each complete ablation to avoid stacking of ablations in the same location.
    - It is recommended to check all rotations of the CARTO™ map to ensure full circumferential co-axial lesions were achieved on all map surfaces.
- In clinical studies with VARIPULSE™ (admIRE and inspIRE) the median number of ablations delivered was 23 (69 applications) and 16 (49 applications) respectively. Device safety and effectiveness were evaluated in all treated subjects, 75% of whom had fewer than 28 ablations (84 applications) delivered in the admIRE clinical study and 17 ablations (52 applications) in the inspIRE clinical study. The risk of patient injury including peri-procedural stroke and transient ischemic attack and product malfunction may increase with an increased number of ablations (energy applications) delivered during the procedure.

Irrigation

- During advancement and withdrawal of the catheter through the long guiding sheath, maintain the flow rate at 4 ml/min.
- When the catheter is in the cardiac chamber, maintain the flow rate at 4ml/min.
- For irrigation during ablation refer to Table 1.
- For manual mode, if selecting the 30 ml/min[3] flow rate, the irrigation pump must be manually adjusted prior to ablation.

Removal and Reinsertion of the Catheter

1. Turn the Rocker Lever to neutral position to ensure that the tip section is not deflected.
2. With the Rotation Contraction Knob on the handle facing the user, turn the Rotating Contraction Knob counterclockwise until the maximum diameter of the loop is reached.
3. Ensure that the long guiding sheath is not deflected (is in a neutral position).
4. Maintain the irrigation rate at 4ml/min before the withdrawal of the catheter through the long guiding sheath.
5. Applying gentle and slow force, remove the catheter from the long guiding sheath.
6. If coagulum is present, dampen a sterile gauze pad with sterile saline. Gently wipe the catheter tip section with the pad. To avoid damage, do not scrub or twist the catheter loop or the electrodes. Before reinserting the catheter, verify patency. Hold the catheter with the loop facing downward and parallel to the floor. Flush the catheter and verify that saline flows from each electrode.

   The catheter tip may be cleaned as described above for reuse in the same procedure only. Do not resterilize and reuse the catheter.

   WARNING: Do not use the catheter if any electrode remains occluded or if the catheter does not appear to function properly. Dispose of the contaminated catheter as described in the “Disposal” section of this document.
7. Hold the catheter with the loop facing upward and use standard aseptic technique to flush air bubbles from the catheter and tubing.
8. The catheter may now be reinserted in the patient as described earlier in this document.

ADVERSE EVENTS

NOTE: The adverse events in the following summary were observed in clinical studies involving only the use of the VARIPULSE™ Catheter system.

Clinical trial for the VARIPULSE™ Catheter System– Pivotal Study (admIRE)
Of the 274 subjects in the mITT Analysis Set, 9 primary adverse events (AEs) were reported in 8 subjects. See the "Summary of Clinical Studies Conducted for the VARIPULSE™ Catheter system" section below for a complete description of the AEs encountered during the study.

Summary of Clinical Studies – Pivotal Study (admIRE)
The clinical testing described below was performed under IDE with the VARIPULSE™ Catheter system.

SUMMARY OF CLINICAL STUDIES CONDUCTED FOR THE VARIPULSE™ CATHETER SYSTEM

A. Objective

The objective of this study (admIRE) was to demonstrate safety and 12-month effectiveness of the VARIPULSE™ Catheter System for pulmonary vein isolation (PVI) in the treatment of subjects with symptomatic paroxysmal atrial fibrillation (PAF).

B. Study Design

The admIRE study was a prospective, non-randomized, multicenter, clinical investigation conducted at 33 investigational sites in the US. A total of 384 subjects were enrolled in the Pilot Phase (22 subjects) and Pivotal Phase (362 subjects). The Pivotal Phase consisted of the roll-in and main phase. The first 0-2 subjects for each ablating physician using the VARIPULSE™ Catheter System was prospectively assigned as roll-in subjects to verify consistent workflow for study device components and to minimize any learning curve effects.

B.1 – Study Endpoints

The endpoints for the study were as follows:

Primary Safety Endpoint

The primary safety endpoint was the incidence of primary adverse events occurring within 7 days of an ablation procedure which used the VARIPULSE™ Catheter and TRUPULSE™ Generator per protocol.

Primary adverse events included the following adverse events:

- Death (device or procedure related)
- Atrio-Esophageal Fistula*
- Cardiac Tamponade** / Perforation**
- Myocardial Infarction
- Stroke / Cerebrovascular Accident
- Thromboembolism
- Transient Ischemic Attack
- Phrenic Nerve Injury / Diaphragmatic Paralysis
- Heart Block
- Pulmonary Vein Stenosis+
- Pulmonary Edema (Respiratory Insufficiency)
- Vagal Nerve Injury / Gastroparesis
- Pericarditis
- Major Vascular Access Complication / Bleeding

* Atrio-esophageal fistula that occurred greater than one week (7 days) post-procedure and up to 90 days post-procedure was considered and analyzed as a PAE.

** Cardiac Tamponade/Perforation occurring within 30 days of the AF ablation process was considered a PAE

+ Pulmonary Vein Stenosis was considered as a PAE if it occurred anytime during the 12-month follow up period.

Primary Effectiveness Endpoint

The primary effectiveness endpoint was freedom from electrocardiographically documented atrial fibrillation, atrial tachycardia or atrial flutter (AF/AT/AFL) episodes during the effectiveness evaluation period (Day 91-365), and freedom from the following failure modes.

- Acute procedural
  - Failure to confirm entrance block in all pulmonary veins at the end of the procedure.
- Repeat ablation
  - More than 1 repeat ablation procedure for AF/AT/AFL during the 3-Month Blanking Period (Day 0-90 post index procedure).
  - Any repeat ablation procedure for AF/AT/AFL during the evaluation period.
- Non-study Catheter
  - Use of a non-study catheter (NSC) to treat pulmonary vein targets to achieve isolation of clinically relevant PVs and/or to ablate left atrial non-PV AF targets during the index procedure.
  - Use of a non-study catheter to treat pulmonary vein targets to achieve isolation of clinically relevant PVs during repeat procedure in the blanking period.
- Anti-arrhythmic Drug
  - Taking a new AAD (Class I & III) for AF/AT/AFL or taking a previously failed Class I/III AAD at a greater than the highest ineffective historical dose for AF/AT/AFL beyond the 3-month follow-up visit window.
- Continuous AF/AT/AFL
  - Continuous AF/AT/AFL on a standard 12-lead ECG during the effectiveness evaluation period.
- DC Cardioversion
  - Any DC cardioversion procedure during the evaluation period for documented for AF/AT/AFL recurrences.

Secondary Effectiveness Endpoints

- QOL (AFEQT) Improvement:
  - Improvement in total score of Atrial Fibrillation Effect on Quality-of-Life Questionnaire (AFEQT) at 12M post procedure compared to baseline score.

Additional Safety Endpoints

- Incidence of Unanticipated Adverse Device Effects (UADEs)
- Incidence of Serious Adverse Events (SAEs) within 7 days (early onset), >7 to 30 days (peri-procedural) and >30 days (late onset) of initial ablation
- Incidence of bleeding complication (ISTH definitions): a) major, b) clinically relevant non-major and c) minor bleeding

Additional Effectiveness Endpoints

- Acute Procedural Success defined as the percent of subjects with electrical isolation of all PVs with confirmed entrance block at the end of the procedure.
- Freedom from Documented Atrial Tachyarrhythmia off Class I/III AAD defined as freedom from electrocardiographically documented atrial tachyarrhythmia episodes during the effectiveness evaluation period and off Class I/III AAD beyond the 3-month follow-up visit window.

- 12 Months Single Procedure Treatment Success defined as freedom from electrocardiographically documented symptomatic AF/AFL/AT recurrence during the effectiveness evaluation period following a single index ablation procedure and freedom from all other failure modes, including acute procedural failure, non-study catheter failure, AAD failure, and DCCV failure defined in the primary effectiveness endpoint. Any repeat ablation procedure for AF/AT/AFL during follow-up would result in 12 Months single procedure treatment failure.
- Clinical Success defined as the freedom from documented symptomatic atrial tachyarrhythmia episodes based on electrocardiographic data (≥30 seconds on ECG or sponsor provided CEM or Holter device) during the effectiveness evaluation period (Day 91-Day 365).

B.2 – Pre-specified Performance Goal

The performance goal was prospectively established.

- Safety: Performance Goal = 12.0% upper bound of the 95% CI for the primary AE rate
- Effectiveness= 50% which is indicated as the minimum acceptable success rate at 12 months for a paroxysmal AF population in the 2017 HRS/EHRA/ECAS/APHRS/SOLAECE Expert Consensus Statement on Catheter and Surgical Ablation of AF.

B.3 – Subject Disposition and Accountability

The following definitions were used to determine subject disposition:

Table 2. Subject Disposition

Disposition	Definition
Enrolled Subjects	Subjects who signed the informed consent
Excluded Subjects	Subjects who were enrolled but never underwent insertion of the study catheter
Evaluable Subjects	All enrolled subjects who have the study catheter inserted.
Discontinued Subjects	Evaluable subjects who have the study catheter inserted but do not undergo ablation (i.e., no PFA energy is delivered with the study catheter). Discontinued subjects will remain in follow-up for 3 months post catheter insertion.
Lost to Follow-up Subjects	Subjects who were evaluable, but contact was lost after the most recent follow-up visits (despite 3 documented attempts to contact the subject)
Withdrawn / Early Termination Subjects	Subjects who withdrew consent for study participation or were withdrawn by the investigator or were terminated from the study prior to completion of all follow-up visits
Completed Subjects	Enrolled subjects who have not been excluded, discontinued, lost-to-follow-up or withdrawn/ early terminated prior to the final study visit and have completed the 12-month follow up visit.

Table 3. Subject Accountability

Subject Enrollment	Pilot	Roll-in	Main study
Enrolled Subjects	22	71	291
Excluded Subjects	1	7	14
Study Catheter Inserted	21	64	277
Discontinued Subjects	0	0	1
Death	0	1	0
Withdrawn	1	1	2
Early Termination Subjects	0	0	0
Lost to Follow Up Subjects	0	0	5
Completed Subjects	20	62	269

The following definitions were used to determine the analysis set:

Pilot Phase:

- **Safety Analysis (SA) Set**: The Safety Analysis Set is comprised of all subjects enrolled in the pilot phase who had insertion of the study catheter (with or without delivery of PFA).

Pivotal Phase:

- **Safety Analysis (SA) Set:** The Safety Analysis Set is comprised of all subjects enrolled in the pivotal main phase who had insertion of the study catheter (with or without delivery of PFA).
- **All Treated (AT) Analysis Set:** The All Treated analysis set includes all subjects enrolled in the Pivotal main phase who underwent ablation (PFA) with the VARIPULSE™ Catheter System for study related arrythmia.
- **Modified Intent-To-Treat (mITT) Analysis Set**: The mITT analysis set is comprised of all subjects enrolled in the pivotal main phase who met eligibility criteria and had insertion of the study catheter (with or without delivery of PFA).
- **Per-Protocol (PP) Analysis Set**: The PP analysis set is comprised of all subjects enrolled in the pivotal main phase who satisfied the following criteria:
    - Without protocol deviations that would affect the integrity of the data, including subjects who:
        - Did not meet eligibility criteria after undergoing the index procedure.
        - Were enrolled as drug refractory due to being contraindicated to or having intolerable side-effects to Class I & III anti-arrhythmic drugs (AADs) who then received these after enrollment.
        - Missed all protocol-specified electronic arrhythmia monitoring and were not a primary effectiveness failure.
        - Failed to check for entrance block for each targeted PV.
        - Failed to follow required procedural workflow parameters unless resulting from an investigational device malfunction, including:
            - Delivery of less than 2 or more than 6 complete applications per position
            - Delivery of more than 8 complete additional applications per PV
            - Delivery of more than 8 complete reconnection applications per PV after adenosine/isoproterenol challenge
            - Failure to challenge the PVs with adenosine/isoproterenol
    - Had undergone PFA ablation with the VARIPULSE™ Catheter system.
        - Note: Subjects who had insertion of the study catheter but did not undergo PFA ablation due to investigational device related reasons would be considered as exceptions to this criterion. These subjects would remain in the analysis set and be

considered as having failed the primary effectiveness endpoint (i.e., acute effectiveness failures).

- Were treated for the study-related arrhythmia.

Roll-In Analysis Set: The Roll-In Analysis Set is comprised of all subjects enrolled in the pivotal roll-in phase, who had insertion of the study catheter and PFA delivery with the IRE ablation system.

C. Subject Demographics and Baseline Characteristics

C.1 – Demographics

Table 4. Subject Demographics

Variable	Statistics		
	Pivotal Main Phase Safety Analysis Set, N=277	**Pivotal Main Phase mITT Analysis Set, N=274**	**Pivotal Main Phase, Per-Protocol Analysis Set, N=255**
Age (years)			
n	277	274	255
Mean ± Standard Deviation	61.5 ± 10.30	61.5 ± 10.27	61.6 ± 10.28
Median	64.0	64.0	64.0
Q1 / Q3	57.0 / 68.0	57.0 / 68.0	57.0 / 68.0
Min / Max	19 / 76	19 / 75	19 / 75
Sex, n/N (%)			
Male	178 / 277 (64.3%)	176 / 274 (64.2%)	163 / 255 (63.9%)
Female	99 / 277 (35.7%)	98 / 274 (35.8%)	92 / 255 (36.1%)
Of child-bearing potential	8 / 99 (8.1%)	8 / 98 (8.2%)	7 / 92 (7.6%)
Permanently sterilized [or infertile]	6 / 99 (6.1%)	6 / 98 (6.1%)	6 / 92 (6.5%)
Postmenopausal	85 / 99 (85.9%)	84 / 98 (85.7%)	79 / 92 (85.9%)
Ethnicity, n/N (%)			
Hispanic or Latino	7 / 277 (2.5%)	7 / 274 (2.6%)	6 / 255 (2.4%)
Not Hispanic or Latino	215 / 277 (77.6%)	213 / 274 (77.7%)	196 / 255 (76.9%)
Not Reported	55 / 277 (19.9%)	54 / 274 (19.7%)	53 / 255 (20.8%)
Race, n/N (%)			
American Indian/Alaska Native	4 / 277 (1.4%)	4 / 274 (1.5%)	3 / 255 (1.2%)
Asian	12 / 277 (4.3%)	12 / 274 (4.4%)	11 / 255 (4.3%)
Black or African American	7 / 277 (2.5%)	7 / 274 (2.6%)	6 / 255 (2.4%)
White	228 / 277 (82.3%)	226 / 274 (82.5%)	211 / 255 (82.7%)
Race not reported	25 / 277 (9.0%)	24 / 274 (8.8%)	23 / 255 (9.0%)
BMI (kg/m^2)			
n	277	274	255
Mean	28.74 ± 5.912	28.69 ± 5.850	28.75 ± 5.924
Median	27.90	27.95	28.00
Q1 / Q3	24.80 / 31.90	24.80 / 31.90	24.80 / 32.00
Min / Max	17.6 / 56.6	17.6 / 56.6	17.6 / 56.6

Table 5. Baseline Characteristics

Variable	Statistics		
	Pivotal Main Phase Safety Analysis Set, N=277	**Pivotal Main Phase mITT Analysis Set, N=274**	**Pivotal Main Phase, Per-Protocol Analysis Set, N=255**
Cardiovascular Medical History	209 / 277 (75.5%)	207 / 274 (75.5%)	193 / 255 (75.7%)
Congestive heart failure	9 / 277 (3.2%)	8 / 274 (2.9%)	8 / 255 (3.1%)
Coronary disease	55 / 277 (19.9%)	54 / 274 (19.7%)	49 / 255 (19.2%)
Vascular disease	17 / 277 (6.1%)	17 / 274 (6.2%)	17 / 255 (6.7%)
Myocardial infarction	8 / 277 (2.9%)	8 / 274 (2.9%)	7 / 255 (2.7%)
Hypertension (systemic)	148 / 277 (53.4%)	147 / 274 (53.6%)	136 / 255 (53.3%)
Pulmonary hypertension	6 / 277 (2.2%)	6 / 274 (2.2%)	6 / 255 (2.4%)
Cardiomyopathy	8 / 277 (2.9%)	8 / 274 (2.9%)	8 / 255 (3.1%)
Left ventricular hypertrophy	4 / 277 (1.4%)	4 / 274 (1.5%)	4 / 255 (1.6%)
Significant valvular disease	1 / 277 (0.4%)	1 / 274 (0.4%)	1 / 255 (0.4%)
Hyperlipidemia	135 / 277 (48.7%)	133 / 274 (48.5%)	124 / 255 (48.6%)
Documented Thromboembolic Events	10 / 277 (3.6%)	9 / 274 (3.3%)	8 / 255 (3.1%)
Transient ischemic attack (TIA)	8 / 277 (2.9%)	7 / 274 (2.6%)	6 / 255 (2.4%)
Stroke	1 / 277 (0.4%)	1 / 274 (0.4%)	1 / 255 (0.4%)
Other Medical History [+]	93 / 277 (33.6%)	92 / 274 (33.6%)	86 / 255 (33.7%)
Type II Diabetes	30 / 277 (10.8%)	30 / 274 (10.9%)	29 / 255 (11.4%)
Obstructive sleep apnea (OSA)	77 / 277 (27.8%)	76 / 274 (27.7%)	71 / 255 (27.8%)
OSA treated	58 / 277 (20.9%)	57 / 274 (20.8%)	53 / 255 (20.8%)
CPAP for OSA	48 / 277 (17.3%)	48 / 274 (17.5%)	44 / 255 (17.3%)
CHA_2DS_2-VASc Score	1.7 ± 1.29	1.7 ± 1.26	1.7 ± 1.26
AAD Failed- Class I&III Mean ± SD	1.0 ± 0.22	1.1 ± 0.22	1.0 ± 0.22
Duration of symptomatic paroxysmal AF (months) Mean ± SD	52.69 ± 73.39	52.31 ± 72.828	52.14 ± 73.140
Symptomatic PAF episodes within 12 months	42.9 ± 90.15	43.4 ± 90.54	42.0 ± 90.43
Arrythmia other than paroxysmal AF, n/N (%)			
Atrial Tachycardia Origin Undefined	8 / 277 (2.9%)	8 / 274 (2.9%)	8 / 255 (3.1%)
Atrial Flutter Typical Right	16 / 277 (5.8%)	16 / 274 (5.8%)	15 / 255 (5.9%)
Atrial Flutter Origin Undefined	29 / 277 (10.5%)	29 / 274 (10.6%)	24 / 255 (9.4%)
AV Node Re-Entry Tachycardia (AVNRT)	1 / 277 (0.4%)	1 / 274 (0.4%)	1 / 255 (0.4%)
LA Diameter (mm)	38.08 ± 5.797	38.06 ± 5.807	38.04 ± 5.847
LVEF (%)	60.7 ± 5.95	60.7 ± 5.90	60.5 ± 5.87

C.2 – Procedural Data

Table 6 to Table 9 present the procedural data.

Table 6. Summary of Ablation Procedure Parameters

Variable	Statistics (Mean ± SD)		
	Pivotal Main Phase Safety Analysis Set, N=277	Pivotal Main Phase mITT Analysis Set, N=274	Pivotal Main Phase, Per-Protocol Analysis Set, N=255
Total LA mapping time (min)	7.67 ±5.031	7.67 ± 5.057	7.83 ± 5.044
Total Left Atrium study catheter dwell time (min)	43.83 ± 18.193	44.02 ± 18.192	43.27 ± 17.765
Total procedure time (min)	95.14 ± 37.189	95.23 ± 37.334	94.98 ± 37.448
Total duration of fluoroscopy at end of procedure (min)	9.99 ± 12.199	10.06 ± 12.237	9.63 ± 12.375
Fluid delivered via study catheter (mL)	254.89 ± 131.980	255.53 ± 132.295	252.90 ± 132.858

Table 7. Summary of PFA applications, ablation duration and PFA application time

Variable	Statistics (Mean ± SD)		
	Pivotal Main Phase Treated Subjects Set, N=276	Pivotal Main Phase mITT Analysis Set, N=274	Pivotal Main Phase, Per-Protocol Analysis Set, N=255
Number of Complete PFA Applications	74.1 ± 19.29	74.2 ± 19.36	71.7 ± 15.57
Total Ablation Duration (min)*	36.04 ± 19.730	36.18 ± 19.788	35.14 ± 19.443
Total Complete PFA Application Time (sec)**	17.96 ± 4.674	17.98 ± 4.689	17.36 3.771

*Defined as the total duration from the first PFA application to the last PFA application, including idle time in between.

**Defined as the sum of all complete (100%) PFA application times, excluding idle time when PFA is not applied.

In the All Treated analysis set (a subset of Safety analysis set), on average, 74.1 valid applications (range 48-173 applications) were delivered across all ablation targets (PV and non-PV LA targets). Of the treated subjects, 75% had fewer than 84 applications delivered.

Table 8. Summary of Subject-level PFA applications

Variable	Statistics		
	Pivotal Main Phase All Treated Analysis Set, N=276	Pivotal Main Phase mITT Analysis Set, N=274	Pivotal Main Phase Per Protocol Analysis Set, N=255
Number of Valid PFA Applications			
N	276	273	255
Mean	74.1	74.2	71.7
Standard Deviation	19.29	19.36	15.57
Median	70.0	70.0	69.0
Q1 / Q3	60.0 / 84.5	60.0 / 85.0	60.0 / 84.0
Min / Max	48 / 173	48 / 173	48 / 130

Table 9. Ablation Sites Targeted

Ablation Sites Targeted	Statistics (n/N (%))		
	Pivotal Main Phase Treated Subjects Set, N=276	**Pivotal Main Phase mITT Analysis Set, N=274**	**Pivotal Main Phase, Per-Protocol Analysis Set, N=255**
Total	276 / 276 (100.0%)	273 / 274 (99.6%)	255 / 255 (100.0%)
PVs Only	213 / 276 (77.2%)	210 / 274 (76.6%)	200 / 255 (78.4%)
PVs + Non-PV with VARIPULSE™ Catheter	18 / 276 (6.5%)	18 / 274 (6.6%)	17 / 255 (6.7%)
Posterior Wall- Segmental	16 / 276 (5.8%)	16 / 273 (5.9%)	15 / 255 (5.9%)
Other AF Foci	1 / 276 (0.4%)	1 / 273 (0.4%)	1 / 255 (0.4%)
Other linear lesion	1 / 276 (0.4%)	2 / 273 (0.7%)	2 / 255 (0.8%)

C.3 – Arrhythmia Monitoring Compliance

Composite arrhythmia monitoring compliance was calculated per the protocol required assessment schedule of TTM/CEM, ECG and HM. Composite arrhythmia monitoring compliance was **99.2%** at 6-Month and **96.8%** at 12-Month post-procedure.

C.4 – Adverse Events (AE)

C.4.1. Primary Safety Endpoint – Primary Adverse Events

The primary safety endpoint for this study was defined as the incidence of early onset (within 7 days of ablation procedure) Primary AEs for subjects undergoing a study ablation procedure. Per study protocol, the primary safety analysis was based on the primary safety event rate in the Pivotal Phase Main mITT analysis set (N=274) as the primary analysis population.

Table 10 displays a summary for the primary safety endpoint in subjects with non-missing outcomes in the mITT analysis set. The PAE rate was 2.9% and the exact one-sided 97.5% upper confidence bound was 5.7%, which is less than the pre-specified performance goal of 12%. Therefore, the results indicate that the study met the pre-defined performance goal for the primary safety endpoint.

The primary safety analysis was repeated in the Pivotal Main Safety analysis set (i.e., all pivotal main phase subjects with the study catheter inserted) as a sensitivity analysis. Similar to the mITT analysis set, a total of eight (8) subjects experienced nine (9) PAEs in the Safety analysis set. There were no additional PAEs observed compared to the mITT analysis set. For Pivotal Main Phase subjects in the Safety analysis set, the PAE rate remained at 2.9% and the exact one-sided 97.5% upper confidence bound was 5.7%, which maintains below the pre-specified performance goal of 12%.

The PAEs reported in this study were cardiac tamponade/perforation (n = 3), major vascular access complication/bleeding (n = 2), pericarditis (n = 1), stroke/cerebrovascular accident (n = 2) and transient ischemic attack (n = 1).

There were NO incidents of device or procedure-related death, atrio-esophageal fistula, myocardial infarction, thromboembolism, phrenic nerve injury/diaphragmatic paralysis, heart block, pulmonary vein stenosis, pulmonary edema, or vagal nerve injury/gastroparesis.

Table 10. Primary Safety Endpoint (Pivotal Main Phase mITT Analysis Set, N=274)

Variable	Number of Subjects with Events	Number of Events	Event Rate n/N (%) [1]	Exact One-Sided 97.5% Upper Confidence Bound
Primary Adverse Event	8	9	8 /272 (2.9%)	5.7%
Type of Primary Adverse Event				
Cardiac Tamponade /perforation	3*	3	3 /272 (1.1%)	
Major Vascular Access Complication/ Bleeding	2	2	2 /272 (0.7%)	
Pericarditis	1*	1	1 /272 (0.4%)	
Stroke/Cerebrovascular Accident	2	2	2 /272 (0.7%)	
Transient Ischemic Attack	1	1	1 /272 (0.4%)	

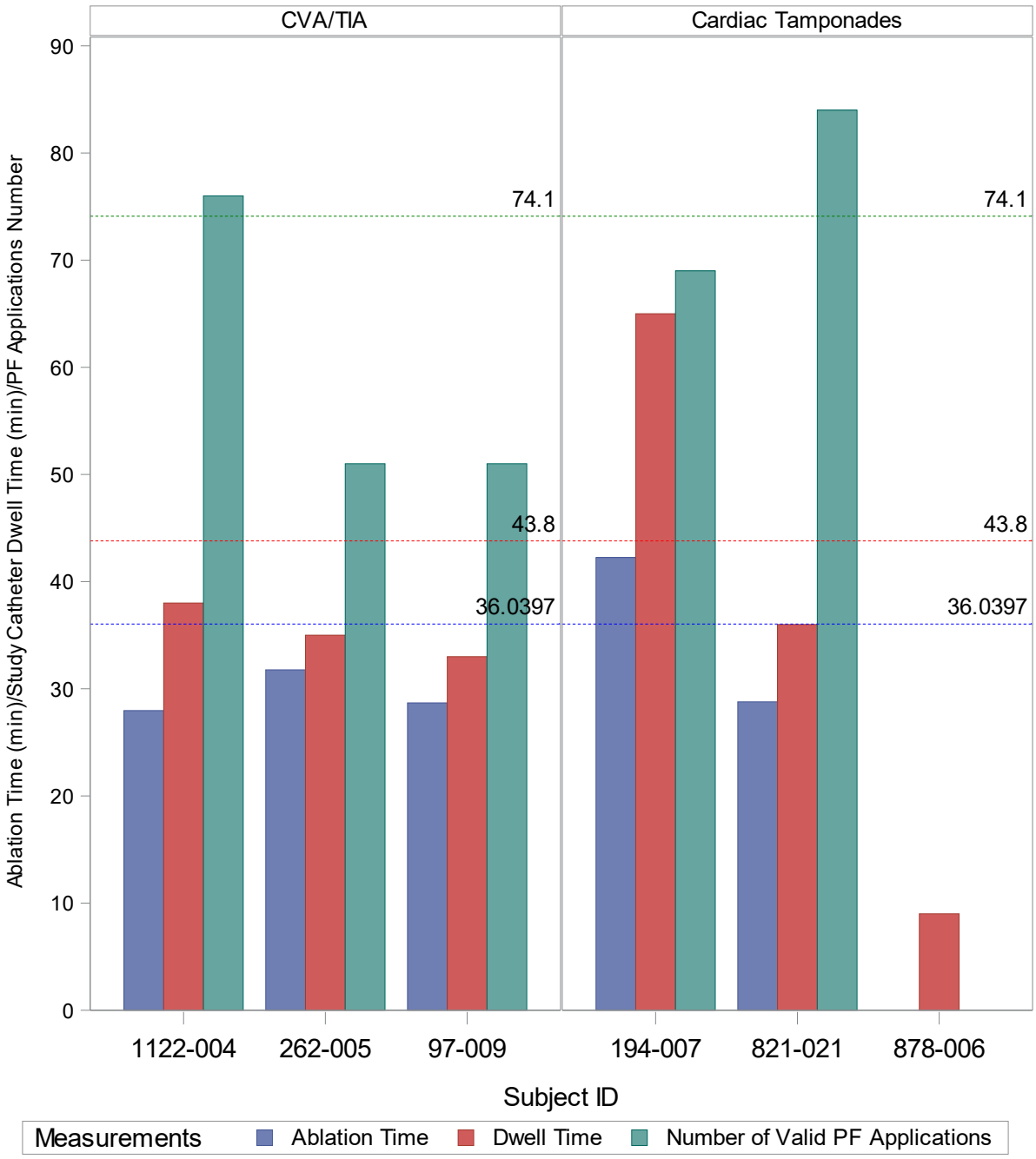
[1] Subjects without at least 3 months of follow-up are excluded from the primary safety analysis unless they experienced a PAE.

*One (1) subject experienced two (2) PAEs: Cardiac tamponade/perforation and Pericarditis.

C.4.2. Ablation Time (min), Study Catheter Dwell Time (min), and Number of PF Applications in subjects experiencing Cardiac Tamponades, CVA/Stroke, or TIA

Figure 1 presents the ablation time, study catheter dwell time, and number of applications for subjects experiencing a cardiac tamponade or CVA (stroke) or TIA as PAE compared to the mean values for subjects in the Pivotal Phase Main Safety analysis set. The procedural workflow in subjects who experienced a PAE appeared to be comparable to that in other subjects in the Pivotal Main Phase Safety analysis set who did not experience any PAE.

Figure 1. Comparison of Ablation Time (min), Study Catheter Dwell Time (min), and Number of PF Applications in subjects experiencing Cardiac Tamponades or CVA/TIA



Note: The dotted line represents the mean of all subjects in the Pivotal Main Phase Safety analysis set.

C.4.3. Serious Non-Primary AEs

Table 11 summarizes the SAEs (by causality and body system) occurring within 30 days of a study ablation procedure that were not classified as Primary AEs by protocol definition.

Table 11. Serious Non-Primary AEs (Occurring 0-30 Days Post Index/Repeat Ablation Procedure) by Severity, Causality and Outcome (Pivotal Main Phase Safety Analysis Set, N=277)

Severity	Relationship to VARIPULSE™ Catheter	Relationship to TRUPULSE™ Generator	Relationship to Procedure	Outcome	AE Term
0-7 days Post Ablation Procedure					
Severe	Causal Relationship to Device	Causal Relationship to Device	Causal Relationship to Procedure	Recovered/Resolved	Cardiac arrest
Moderate	Not Device Related	Not Device Related	Causal Relationship to Procedure	Recovered/Resolved	Urinary retention
	Not Device Related	Not Device Related	Possible Procedure Related	Recovered/Resolved	Migraine
	Not Device Related	Not Device Related	Not Procedure Related	Recovered/Resolved	Atrial fibrillation
	Not Device Related	Not Device Related	Not Procedure Related	Recovered/Resolved	Diverticulitis
	Not Device Related	Not Device Related	Not Procedure Related	Recovered/Resolved	Pericardial effusion
	Not Device Related	Not Device Related	Not Procedure Related	Recovered/Resolved	Urinary retention
	Not Device Related	Not Device Related	Not Procedure Related	Recovered/Resolved	Wound secretion
	Not Device Related	Not Device Related	Not Procedure Related	Recovered/Resolved with sequelae	Pneumonia
Mild	Not Device Related	Not Device Related	Not Procedure Related	Recovered/Resolved	Lower gastrointestinal haemorrhage
8-30 days Post Ablation Procedure					
Mild	Possible Device Related	Possible Device Related	Probable Procedure Related	Recovered/Resolved	Pleural effusion

C.4.4. Subject Deaths

One roll-in subject died of out-of-hospital cardiac arrest with a suspected diagnosis of acute myocardial infarction on day 39 post study procedure without immediate complications or significant findings on 7-day and 1-month follow-up evaluation. No autopsy was performed. The death was classified as not related to either the study device(s) or procedure by the investigator. .

C.4.5. Primary Safety by Sex

Table 12 summarizes the primary safety endpoint by sex for subjects in the Pivotal Main Phase mITT analysis set. The difference in PAE rates between male and female subjects was significant at the 0.15 alpha level (p-value=0.027 < 0.15).

Table 12. Primary Safety Endpoint by Sex
(Pivotal Main Phase mITT Analysis Primary Set, N=274)

Primary Safety Endpoint	Male	Female	P-value [1]
n/N	2 / 174	6 / 98	0.027
% [1]	1.1%	6.1%	
95% Confidence Interval	(0.1%, 4.1%)	(2.3%, 12.9%)	

[1] By Fisher's exact test.

Further analysis of the device related PAEs by sex narrowed the gap in sex-based difference in PAE rates to 1.1% in male subjects and 4.1% in female subjects, with a 95% UCB of 4.1% in male subjects and 10.1% in female subjects.

Table 13 summarizes the primary safety endpoint related to the study devices (VARIPULSE™ Catheter and/or TRUPULSE™ Generator) by sex for subjects in the Pivotal Main Phase mITT analysis set. The difference between sex narrowed. The device related PAE rates was 1.1% in male subjects and 4.1% in female subjects.

Table 13. Primary Safety Endpoint Related to Study Device(s) by Sex
(Pivotal Main Phase mITT Analysis Set, N=274)

Primary Safety Endpoint	Male	Female
n/N	2 / 174	4 / 98
%	1.1%	4.1%

C.5 – Effectiveness Endpoint

C.5.1. Primary Effectiveness Endpoint

Primary Analysis

Per study protocol, the primary effectiveness analysis was based on primary effectiveness success using the Per Protocol Analysis set as the primary analysis population. As shown in Table 14, the primary effectiveness success rate in the Pivotal Main Per Protocol Analysis set was 74.4% and the exact one-sided 97.5% lower confidence bound was 68.5%, which is greater than the pre-specified performance goal of 50%. Therefore, the results indicate that the study met the primary effectiveness endpoint.

Table 14. Primary Effectiveness Endpoint
(Pivotal Main Phase Per Protocol Analysis Set, N=255)

Outcome	Number of Subjects	Success Rate n/N (%)	Exact One-Sided 97.5% Lower Confidence Bound
Success	183	183/246 (74.4%)	68.5%
Failure	63		
Missing	9		

Kaplan-Meier freedom from effectiveness failure for subjects (Figure 2) in the Pivotal Main Phase Per Protocol Analysis set indicates a success rate of 74.6% at 12 months.

Figure 2. Kaplan-Meier Analysis of Time to Primary Effectiveness Failure [1]
(Pivotal Main Phase, Phase Per Protocol Analysis Set, N=255)

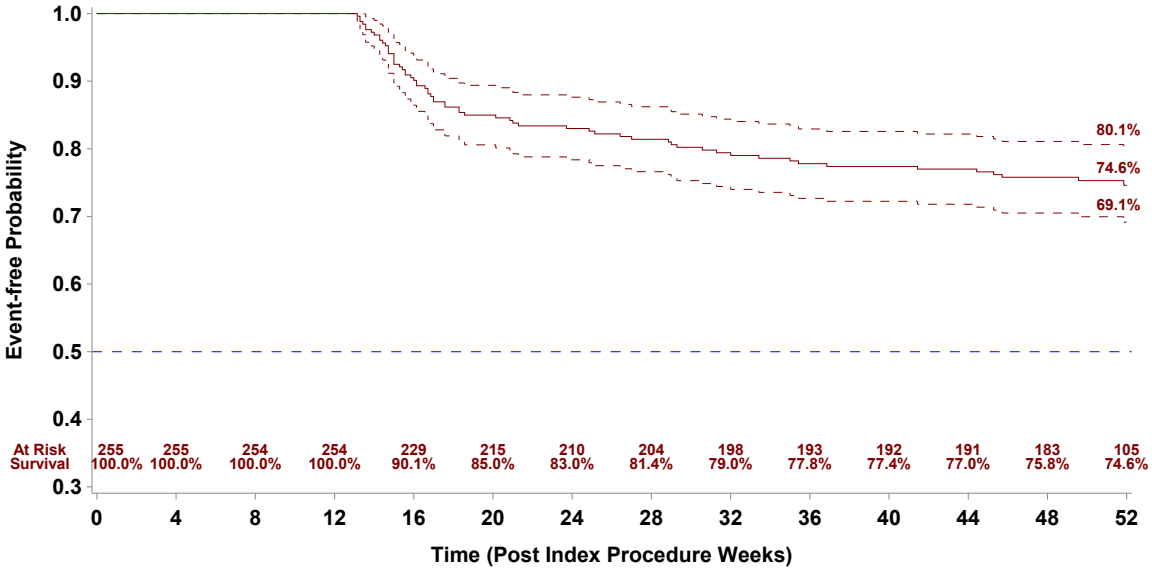


Table 15 summarizes the first failure reason for subjects in the Pivotal Main Phase Per Protocol analysis set. For first failures, if a subject had more than one failure event, only the earliest failure event was summarized. For total failures, if a subject had failed for multiple modes, the subject was counted in all applicable failure modes.

The first failure events reported among 63 primary effectiveness failures included documented recurrence of atrial tachyarrhythmias (60/63) and new AAD failure (3/63). Among the 89 total failure events reported, 61 were documented recurrence of atrial tachyarrhythmias, 19 repeat ablation, and 8 AAD failures.

Table 15. Primary Effectiveness Failures (Pivotal Main Phase Per Protocol Analysis Set, N=255)

Outcome	Number of Subjects
First Failures [1]	63
Acute Procedural Failure [2]	0
Recurrence [3]	60
Atrial Fibrillation	47
Atrial Flutter	11
Atrial Tachycardia	2
Repeat Ablation Failure [4]	0
NSC Failure [5]	0
AAD Failure [6]	3
AAD Failure - New Drug	3
DCCV Failure [7]	0

[1] First failures: if a subject has at least 1 failure event, only the earliest failure event is considered and summarized by its failure mode. Total Failures: if a subject has failed for multiple modes, the subject will be counted in all applicable failure modes.
[2] Acute procedural failure is defined as failure to confirm entrance block in all PVs except those that are silent and/or cannot be cannulated post-procedure, or failure to have PFA delivery with the study catheter due to IRE system malfunctions.
[3] Recurrence is defined as documented (symptomatic and asymptomatic) atrial tachyarrhythmia (AF, AT or AFL of unknown origin) episodes based on electrocardiographic data (>=30 seconds on ECG or sponsor provided CEM or Holter device) during the effectiveness evaluation period.
[4] Repeat ablation failure is defined as more than 1 repeat ablation procedure for AF/AT/ AFL of unknown origin during the blanking period, or any repeat ablation procedure for AF/AT/ AFL of unknown origin during the evaluation period.
[5] NSC failure is defined as use of a NSC to treat PV targets to achieve isolation of clinically relevant PVs (all PVs except those that are silent and/or cannot be cannulated) and/or ablation of left atrial non-PV AF targets during the index procedure, or use of a NSC to treat PV targets to achieve isolation of clinically relevant PVs (all PVs except those that are silent and/or cannot be cannulated) during a repeat procedure in the blanking period.
[6] AAD failure is defined as taking a new AAD (Class I and III) for atrial tachyarrhythmia (AF, AT or AFL of unknown origin), or a previously failed AAD at a greater than the highest ineffective historical dose for AF/AFL/AT during the effectiveness evaluation period.
[7] DCCV failure is defined as Any DCCV procedure during the effectiveness evaluation period for documented atrial tachyarrhythmia recurrences ascertained through protocol specified arrhythmia monitoring methods (12 lead ECG, CEM and Holter).

Sensitivity Analysis

The analysis of primary effectiveness endpoint was repeated in all Pivotal Main Phase mITT and All Treated analysis sets as a sensitivity analysis. The success rate was 72.7% (Table 16) and the exact one-sided 97.5% lower confidence bound was 66.9%, which is also greater than the pre-specified performance goal of 50% (Table 16). The results suggested that the success rates in all three analysis cohorts met the effectiveness performance goal.

Table 16. Primary Effectiveness Endpoint

Pivotal Main Phase	Outcome	Number of Subjects	Success Rate n/N (%)[1]	Exact One-Sided 97.5% Lower Confidence Bound
mITT (N=274)	Success	192	192/264 (72.7%)	66.9%
	Failure	72		
	Missing [2]	10		
All Treated (N=276)	Success	194	194/267 (72.7%)	66.9%
	Failure	73		
	Missing [2]	9		

[1] N represents the number of subjects who either 1) experienced a primary effectiveness failure, 2) completed 12-months of follow-up, or 3) had ECG, CEM, or HM data beyond 335 days after the index procedure.
[2] Subjects who did not experience a primary effectiveness failure and did not complete 12-months of follow-up or have ECG, CEM, or HM data beyond 335 days of the index procedure.

C.5.2. Primary Effectiveness by Sex

Table 17 presents the primary effectiveness outcome for the Pivotal Main Phase Per-Protocol Analysis set by sex. The difference in primary effectiveness success rates between males and females was significant at the 0.15 alpha level (p-value=0.095 < 0.15).

Table 17. Primary Effectiveness Endpoint by Sex (Pivotal Main Phase Per Protocol Analysis Set, N=255)

Primary Effectiveness Endpoint Success	Male	Female	P-value [1]
n/N	**122 / 156**	**61 / 90**	**0.095**
%	**78.2%**	**67.8%**	

[1] By Fisher's exact test.

C.6 – Secondary Effectiveness Endpoint

C.6.1 Change in Overall AFEQT Score

Table 18 presents the change in overall AFEQT scores at 12-months compared to baseline. There was a significant improvement in AFEQT score at 12-months compared to baseline for subjects in the Per Protocol Analysis set (p<0.001) and the mITT Analysis set (p<0.001).

Table 18. Change in Overall AFEQT Score at 12 Months

	Pivotal Main Phase Per Protocol Analysis Set (N=255)		Pivotal Main Phase mITT Analysis Set (N=274)
AFEQT Questionnaire	**Statistics**	**P-value [1]**	**Statistics**
Change from Baseline			
Mean	32.033	<.0001	32.077
95% Confidence Interval	(29.092, 34.973)		(29.247, 34.906)

[1] By paired t-test

C.7 – Additional Effectiveness Endpoint

C.7.1 Acute Procedural Success

Table 19 presents the acute procedural success (confirmation of entrance block in all PVs) among subjects treated with the study catheter in the Per Protocol and mITT Analysis sets. Acute procedural success was defined as confirmation of entrance block in all PVs. One (1) subject in the mITT analysis set who was discontinued due to non-PFA system related reasons prior to PFA delivery was treated as missing. All subjects that were treated (i.e., energy delivered) with the study catheter achieved acute procedural success.

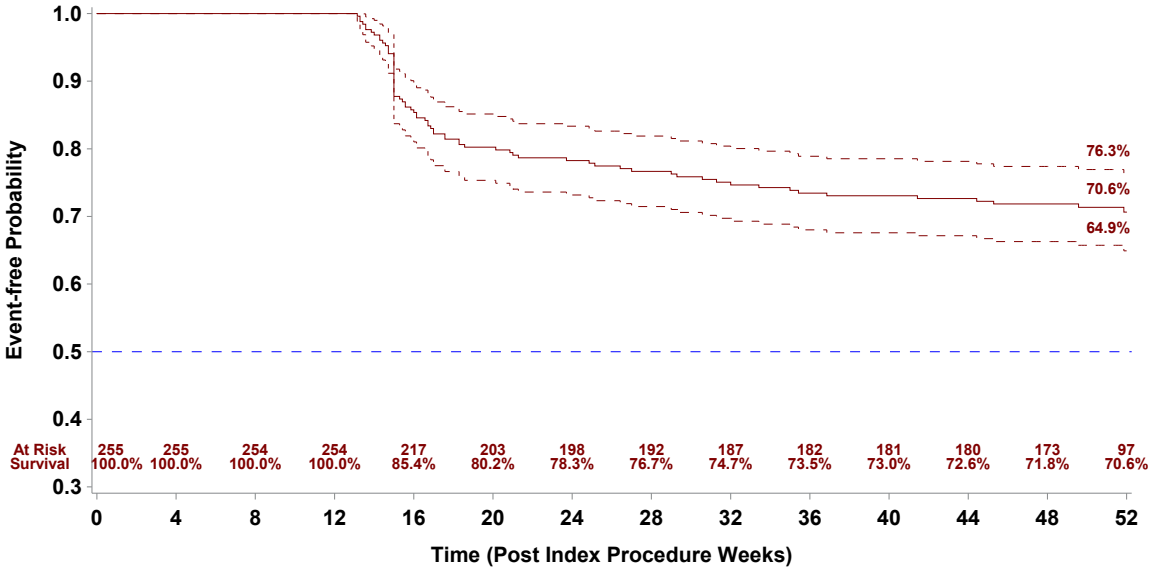
Table 19. Acute Procedural Success

Pivotal Main Phase	Number of Subjects	Number of Successful Subjects	Success Rate n/N (%)
Per Protocol Analysis Set (N=255)	255	255	255 / 255 (100.0%)
MITT Analysis Set (N=274)	273	273	273/273 (100.0%)

C.7.2 Freedom from documented atrial tachyarrhythmia off Class I/III AAD

Kaplan-Meier estimates (Figure 3) used to characterize the time to first documented (symptomatic and asymptomatic) atrial tachyarrhythmia episodes off Class I/III AAD was 70.6% in the Pivotal Main Phase Per Protocol analysis set.

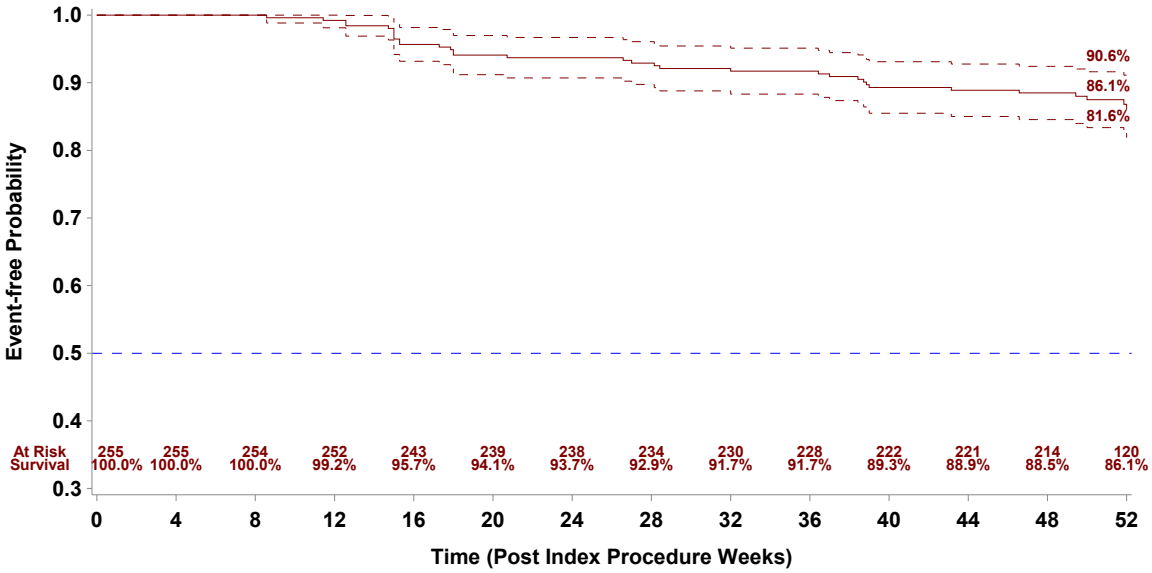
Figure 3. Kaplan-Meier Analysis of time to Documented Recurrence and Off AAD (Pivotal Main Phase Per Protocol Analysis Set, N = 255)



C.7.3 12 Month Single Procedure Treatment Success

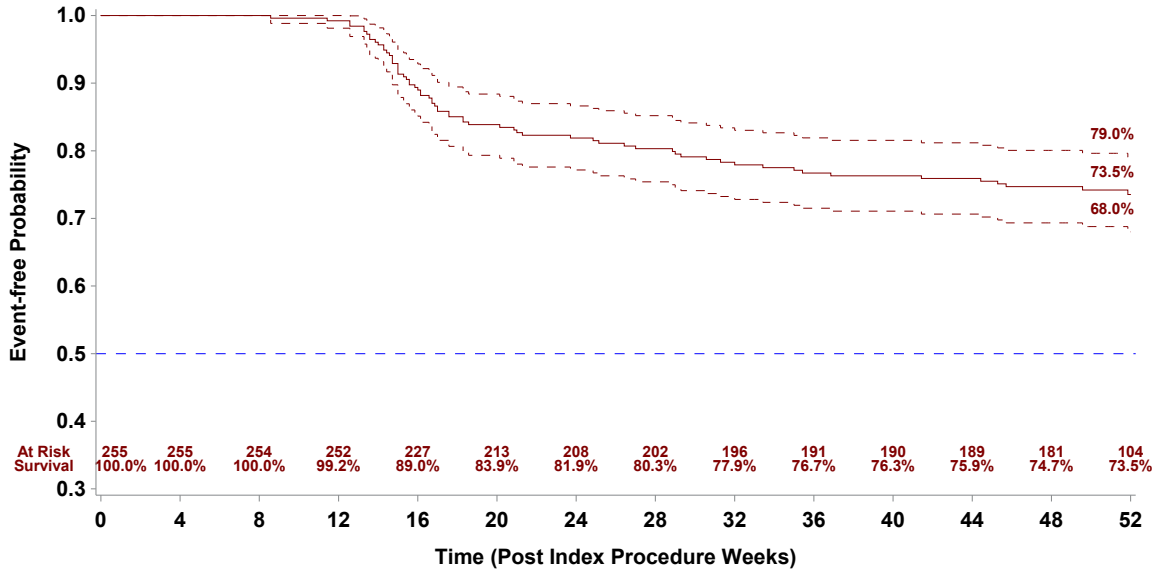
The 12-Month single procedure success rate was 86.1% (Figure 4) in the Pivotal Main Phase Per-Protocol analysis set. Similar results were reported for the mITT analysis set.

Figure 4. Kaplan-Meier Analysis of Single Procedure Treatment Success (Pivotal Main Phase Per Protocol Analysis Set, N = 255)



An ad hoc analysis was performed to assess the 12-Month single procedure success, also counting asymptomatic recurrences in the effectiveness evaluation period as failures. As shown in Figure 5 below, the 12-month single procedure success rate taking both asymptomatic and symptomatic recurrences into account was 73.5% in the Pivotal Main Phase Per Protocol analysis set. Similar results were reported for the mITT analysis set.

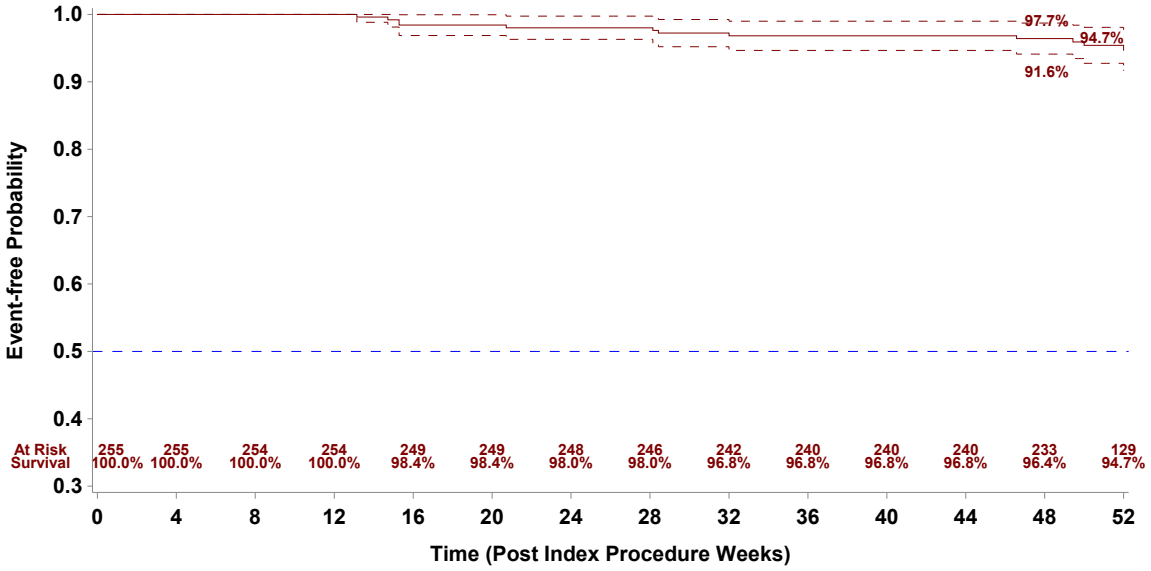
Figure 5. Kaplan-Meier Analysis of Modified Single Procedure Treatment Success (Pivotal Main Phase Per Protocol Analysis Set, N = 255)



C.7.4. Clinical Success

Kaplan-Meier estimates (Figure 6) used to characterize the time to first documented symptomatic atrial tachyarrhythmia episodes based on electrocardiographic data (≥30 seconds on ECG or sponsor provided CEM or Holter device) during the effectiveness evaluation period (Day 91-Day 365) in the Pivotal Main Phase Per Protocol analysis set. The categorization of “symptomatic vs. asymptomatic” was documented based on subject-reported outcomes. The 12-month clinical success was 94.7% (95% CI: 91.6%, 97.7%) in the Per Protocol analysis set.

Figure 6. Kaplan-Meier Analysis of Clinical Success
(Pivotal Main Phase, Per Protocol Analysis Set, N=255)



D. Study Conclusion

The results of the admIRE pivotal study demonstrated the safety and effectiveness of the VARIPULSE™ Catheter System for PV isolation to treat patients with symptomatic drug-refractory PAF .

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